Acousto-Cymatic Phase-Change Memory: A Speculative Theoretical Framework for High-Density Data Storage in Confined Aqueous Media

Sahar Barak*1

1Independent Researcher

Preprint — June 6, 2026

Speculative "idea paper" (cf. `physics.pop-ph`/ `cs.ET`).

Abstract

As conventional magnetic and solid-state storage technologies approach fundamental physical scaling limits—most notably the superparamagnetic limit [1, 2]—new architectures are sought to meet global data demand. This paper proposes, as a deliberately speculative thought experiment, a data-storage architecture we call *Acousto-Cymatic Phase-Change Memory* (ACPCM). By drawing engineering analogies from Heat-Assisted Magnetic Recording (HAMR) [3], three-dimensional charge-trap flash [5, 6], megahertz acoustic holography [12], and scientific data sonification [25], we sketch a method for encoding digital data into the physical topology of a confined, rapidly frozen aqueous "pod." In the proposal, data is encoded as acoustic frequencies, organized into a three-dimensional grid by cymatic pressure nodes, and locked into a solid lattice by a HAMR-style "broadcast-and-freeze" phase transition; read-back is designed around established megahertz acoustic holography and neural-network field inversion, isolating the speculative burden on the write side. We are explicit throughout about which ingredients are established physics and which are contested or non-mainstream ("water memory," structured-water chains, and W. Russell's octave cosmology). Rather than asserting feasibility, we cast the architecture as a set of falsifiable predictions and identify the physical obstacles—above all the energetics, lattice distortion, and the ~50 fs memory loss and picosecond reorganization of the hydrogen-bond network [29]—that would have to be overcome for any such scheme to be viable. The contribution is a conceptual synthesis and a research agenda, *not* a validated device.

Keywords: unconventional computing; acoustic holography; acoustofluidics; phase-change media; ice nucleation; data sonification; speculative storage architectures.

This is a **speculative theoretical proposal**, written in the spirit of an "idea paper." It combines *established* engineering mechanisms with *contested or refuted* premises, and it should be read as a hypothesis generator, not as a report of a working device or of accepted science.

- **Established** (well supported in the literature): the superparamagnetic limit and HAMR [1, 2, 3, 4]; 3D charge-trap flash [5, 6, 7]; chalcogenide phase-change memory [8, 9, 10]; passive acoustic-hologram phase plates and acoustofluidic patterning [12, 17, 20]; deep-learning acoustic-hologram synthesis [23]; and data sonification as a methodology [24, 25].

*Correspondence: `sahar.h.barak@gmail.com`. Authorship and affiliation are placeholders to be edited before any submission or circulation.

- **Contested / minority view**: "exclusion-zone / fourth-phase" structured water [33] (the interfacial effect is real but its structural interpretation is disputed [32]); quantum coherence in microtubules (Orch-OR) [30], challenged on decoherence grounds [31].
- **Refuted / pseudoscientific** (used here only as historical or motivational framing, *not* as evidence): "water memory" [34], which an on-site investigation found did not survive blinding [35] and which later revivals [36] did not reproduce; and W. Russell's octave periodic system [37].

A claim's appearance in this paper is *not* an endorsement of its validity. Crucially, the established citations ground only their own subsystems; none of them lends support to the central acousto-cymatic conjecture, which is ours and is clearly labelled as speculation throughout. Section 8 and table 1 make the dependencies explicit.

1 Introduction

The data-storage industry is constrained by a set of competing requirements—high areal density (which demands small grains), thermal (bit) stability (which demands high magnetic anisotropy), and writeability (which demands a switching field a head can actually supply)—that become mutually exclusive as bits shrink; in the later literature this is often called the *recording trilemma* [2]. The underlying physics is thermodynamic: as a grain's volume V shrinks, its anisotropy energy $K_\mathrm{u}V$ falls toward the thermal energy $k_\mathrm{B}T$, and once the stability ratio $K_\mathrm{u}V/k_\mathrm{B}T$ drops below ~40–60 the magnetization reverses spontaneously—the *superparamagnetic limit*, whose onset Charap, Lu, and He first projected for thin-film media [1] and which Weller and Moser analyzed in its modern form [2].[1] The industry's response, *Heat-Assisted Magnetic Recording* (HAMR), uses a near-field plasmonic transducer to heat a high-anisotropy medium close to its Curie temperature for well under a nanosecond, transiently collapsing its coercivity so a modest field can write a bit, after which rapid cooling "freezes" the magnetization into a thermally stable state [3]; the near-field-transducer write process has been demonstrated experimentally [4] and HAMR is now shipping commercially.

In parallel, solid-state drives escaped planar scaling by going vertical: 3D NAND with *charge-trap flash* (CTF) stores charge in a discrete, electrically insulating silicon-nitride layer rather than a continuous conductive floating gate, so a single local defect leaks only its neighborhood rather than draining a shared pool [5]; the move to stacked, "punch-and-plug" vertical strings decoupled density from lithography [6, 7].

This paper asks a deliberately unconventional question: *what if the two core ideas behind these technologies—rapid state-change writing (HAMR) and microscopic spatial isolation (CTF)—were applied to an exotic medium, namely water frozen on demand?* Liquid water is, on its face, the worst imaginable storage medium: its hydrogen-bond network loses memory of its configuration within ~50 fs and fully reorganizes on a picosecond timescale [29], so any imprint in the *liquid* is erased almost immediately. Our proposal confronts this directly by never trying to store information in the liquid: information exists only fleetingly as a broadcast acoustic field and is committed to memory by an *instantaneous phase change* into ice, in loose analogy to HAMR's heat-then-quench write cycle [3] and to chalcogenide phase-change memory's melt–quench amorphization [8, 9].

We stress at the outset (and repeat in section 8) that several premises in the draft framing of this idea—"water memory," long-lived "structured-water" chains, and an octave-based re-mapping of the periodic table—are *not* part of accepted physics or chemistry; the first was specifically investigated and not reproduced [35, 34]. We include them transparently, label their status, and engineer the architecture so that the refuted ideas carry no logical load. The intended contribution is a clean conceptual synthesis across fields that are rarely discussed together, plus an explicit map of where the idea would break.

[1] Their projected areal-density upper bound of ~36 Gbit/in^2 for longitudinal media was later surpassed by media engineering and perpendicular recording; we cite it as a projected estimate, not a proven wall.

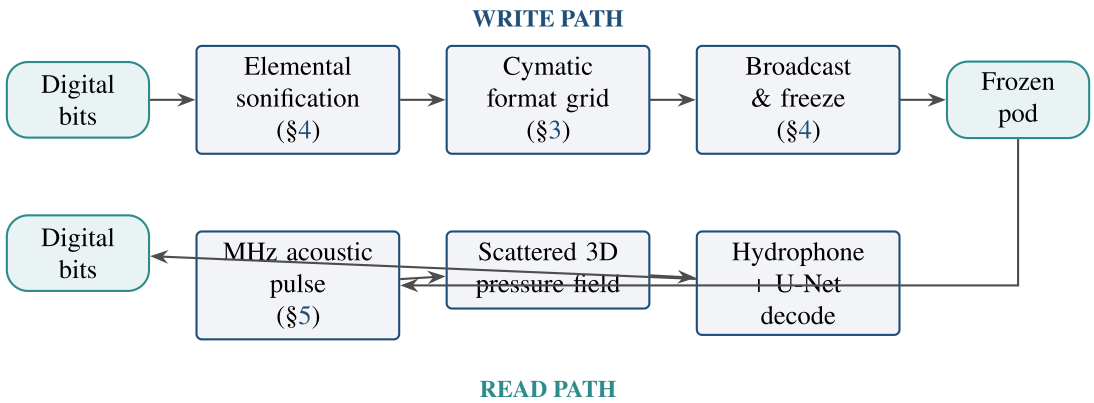

Figure 1. End-to-end ACPCM data path. The write path encodes bits as acoustic tones, imposes a cymatic addressing grid, and commits the field to a solid lattice by freezing. The read path probes the frozen pod with a megahertz pulse, treats the ice topology as an acoustic phase plate, and decodes the scattered field with a neural network. The read path and the write *actuation* reuse established techniques; the speculative weight is isolated in the freeze step (Assumption 3). Each block's epistemic status is given in table 1.

Contributions.

1. A subsystem-by-subsystem architecture (medium, format, write, read) that transposes HAMR and CTF design principles onto a rapidly frozen aqueous medium (sections 2 to 5).
2. An explicit *epistemic ledger* (table 1) separating established mechanisms from speculative ones, so the proposal cannot be mistaken for a validated device and so that mainstream citations are not laundered into supporting fringe claims.
3. A set of falsifiable predictions and order-of-magnitude obstacles (sections 7 and 8) that define what an experimental test would have to measure.

2 The storage medium: confined aqueous "pods"

For water to act even momentarily as an addressable medium, it cannot be a bulk volume. In bulk liquid, distinct driving frequencies superpose and produce beat phenomena, so independently addressed "cells" would modulate and erase one another (crosstalk). The proposal therefore confines the medium to microscopic *pods* that physically isolate addressable regions—an acoustic analogue of CTF's discrete charge traps, where isolation, not bulk capacity, is the design goal [5].

Assumption 1 (Confinement enables isolation)**.** Sub-micron confinement suppresses inter-cell acoustic crosstalk enough that independently addressed regions do not annihilate one another during the write window.

As biological *motivation only*, we note the recurring (and contested) claim that cytoskeletal microtubules—hollow cylinders with an inner diameter of order 15 nm—might shield ordered internal water from thermal noise, a premise of the Orch-OR quantum-biology hypothesis [30]. That premise is itself disputed: Tegmark's decoherence estimates put any quantum coherence in warm, wet microtubules many orders of magnitude below cognitive timescales [31] (a critique that proponents have in turn disputed on modelling grounds). We therefore use microtubules strictly as an intuition pump for "confinement protects order" and assert nothing about biological memory. Likewise, the notion of long-lived spherical "water pearls" or "trimmed chains" as information carriers belongs to the structured-water literature [33]; the underlying interfacial exclusion-zone effect is real and replicated, but its "fourth-phase" structural interpretation is a contested minority view that conventional mechanisms may explain better [32]. In ACPCM such structures, if they exist at all, are relevant only *after* freezing has removed the liquid's thermal lability, not before.

3 Format: a cymatic cylinder–head–sector grid

Conventional drives impose logical structure—tracks and sectors mapped to logical block addresses—onto a physical medium. ACPCM proposes to impose this structure *acoustically*. Using megahertz ultrasound projected into the pod, intersecting wavefronts establish a three-dimensional pattern of standing-wave nodes (pressure minima) and antinodes. The nodal surfaces are intended to act as "sector walls," partitioning the pod into addressable voxels with definite spatial coordinates.

The *patterning of matter by sound* is established physics. Since Faraday it has been known that vibrating surfaces and liquid layers drive particles into ordered figures [11]; modern acoustofluidics uses MHz standing waves to focus, separate, and trap particles and cells at radiation-force nodes [20], down to one cell per well [19], and single-sided holographic arrays form programmable "acoustic tweezers" and vortex traps that levitate and manipulate discrete objects [17, 18]. Passive acoustic-hologram phase plates can sculpt remarkably intricate 3D pressure landscapes from a single transducer [12, 13].

> The *spatial arrangement of discrete particles, droplets, or cells at acoustic nodes* is established and reproducible (fig. 2)—but it is transient (it lasts only while the field is on) and it organizes *pre-existing objects*. The stronger claim that such fields coerce *water molecules themselves* into a persistent "supramolecular network"—sometimes labelled "molecular cymatics"—is **not** established: it has no accepted mechanism and no primary literature specifically linking acoustic-field forcing to supramolecular water ordering (the structured-water literature [33, 32] concerns interfacial proximity, not acoustic induction), and liquid-water hydrogen bonds relax in picoseconds [29]. It is treated here purely as a hypothesis (Assumption 2), and must not be confused with the established assembly result.

Assumption 2 (Cymatic addressing)**.** A megahertz standing-wave field can define stable, reproducible nodal boundaries at the length scale of the intended voxels, and these boundaries survive the freezing transition well enough to serve as a read-time coordinate system.

4 Encoding and writing: the "broadcast-and-freeze" mechanism

4.1 Encoding by elemental sonification

Digital payloads are mapped to acoustic spectra. *Data sonification*—the principled mapping of data to non-speech sound—is a mature methodology with an NSF-commissioned status report [24], a canonical handbook [25], and a formal taxonomy whose defining criterion is that the sound reflect *objective* properties of the data through a systematic, reproducible transform [26]. Mapping the wavelengths of atomic emission lines into the audio band is a documented technique used for analysis, accessibility, and education—e.g. the audification of astronomical spectra [27] and an interactive sonified periodic table [28]. ACPCM uses such a mapping as a codebook: payload symbols index a dictionary of multi-tone acoustic "words." These audio-band tones serve as *codebook symbols*, not literal drive frequencies: the transducer operates in the MHz band (section 3), and each symbol indexes a distinct multi-tone pattern within it.

It is essential to be precise about what sonification *is not*. The light-to-audio mapping (from $\sim 10^{14}$ Hz to $\sim 10^2$–10^4 Hz) is an arbitrary, reproducible rescaling chosen for human perceptibility [28]; it is *not* a physical resonance, it reveals no "hidden physics," and it carries no information beyond what is already in the source

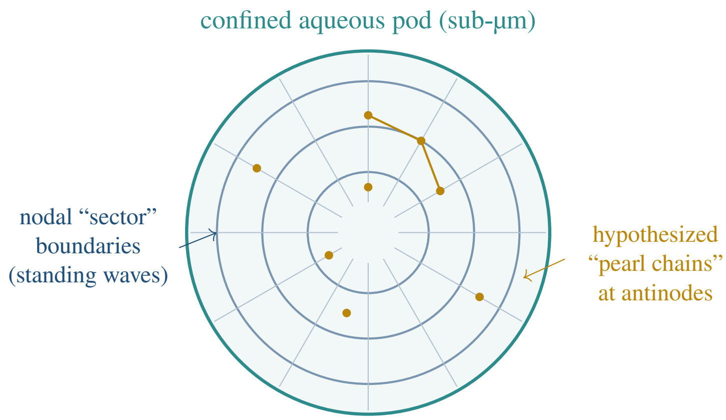

Figure 2. Schematic of a pod with a cymatic cylinder–head–sector (CHS) analogue. Concentric and radial standing-wave nodes (blue) partition the pod into addressable voxels—the established, but transient, particle-patterning physics. The hypothesized “pearl chains” resident at antinodes (amber) belong to the contested structured-water premise (Assumption 2). Not to scale.

data [26]. The draft framing additionally invokes W. Russell’s octave cosmology, in which the elements are arranged on a spiral of ten octaves balancing “centripetal” and “centrifugal” tendencies [37]. We are explicit that this is a **historical/esoteric** system—metaphysics, not chemistry—used here, if at all, only as one arbitrary-but-fixed codebook among many. Nothing in the architecture depends on Russell’s metaphysics being correct; any injective, noise-separated frequency code would serve, and a *designed* code (e.g. orthogonal frequency-division multiplexing) is strictly preferable.

4.2 Writing by phase change

The pod is held near its freezing point. The transducer simultaneously broadcasts (i) the cymatic formatting field of section 3 and (ii) the data tones. While the field is held, the pod undergoes an *instantaneous phase transition* to a solid. In analogy to the melt–quench programming of chalcogenide phase-change memory [9, 10]—and, more loosely, to HAMR’s heat-and-quench write cycle [3]—in which a pulse melts a bit and rapid cooling freezes it into a distinct solid phase, the rapid solidification is hypothesized to “record” the instantaneous acoustic field as a frozen-in topology—displacements, density modulations, and interface positions locked into the lattice.

Assumption 3 (Faithful commit)**.** Freezing transfers the broadcast acoustic field into a static, readable lattice topology with sufficient fidelity and reproducibility to be decoded, and the transition’s own dynamics (stochastic nucleation, latent-heat release, ∼9% volumetric expansion, dendritic growth) do not destroy the encoded pattern.

Assumption 3 is the architecture’s load-bearing physical claim and its most likely point of failure; we return to it in section 8. We emphasize that the HAMR and PCM citations establish only that *their own* rapid-quench mechanisms work; they provide *no* evidence that an acoustic field can be recorded by freezing water. That bridge is ours, and it is conjecture.

5 Reading: megahertz acoustic holography

Once committed to ice, the data lives in microscopic features of the solid, far below the resolution of any microphone. Read-back is posed as an *acoustic holography* problem [12].

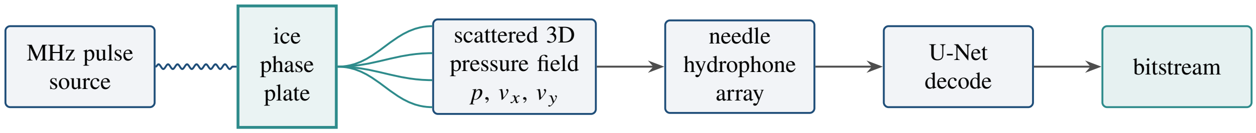

Figure 3. Read chain. A megahertz pulse scatters off the frozen pod, whose topology acts as an acoustic phase plate [12]; the resulting volumetric field (scalar pressure plus, optionally, particle-velocity components) is sampled by a hydrophone array [21, 22] and inverted to a bitstream by a trained network [23]. This read chain reuses only established techniques; the speculative burden lies entirely on the write side (Assumption 3).

The ice as a phase plate. A megahertz ultrasonic pulse illuminates the frozen pod. Because acoustic holography is diffraction-limited, resolving microscopic features requires correspondingly short wavelengths—i.e. MHz (or higher) ultrasound, with millimetre-to-sub-millimetre wavelengths in water and ice [12, 14]. The ice topology then acts as a passive *acoustic phase plate* [12, 15]: the scattered wave is transformed into a complex, three-dimensional volumetric pressure field that encodes the structure it bounced off, and such plates can in principle be stacked or reconfigured to multiplex fields [16].

Capturing the field. The scattered field is sampled by a calibrated needle hydrophone over a measurement surface, the standardized method for spatially mapping ultrasound pressure [21]. We note an honest caveat from the metrology literature: while focal-pressure measurements can be highly *repeatable* (coefficient of variation below 2%), inter-hydrophone *agreement* for tightly focused MHz fields can differ by tens of percent due to spatial averaging and frequency-response uncertainty [22]—a real ceiling on read fidelity. A frequently proposed enhancement is *vectorial* sensing: recording not only the scalar pressure p but the particle-velocity components (v_x, v_y). We flag, however, that in a fluid these are *not* independent degrees of freedom—they are fixed by spatial gradients of the same scalar pressure via Euler's relation—so they add an information *dimension* under a physical constraint rather than free, independent bandwidth; this remains an emerging idea, not a settled gain.

Neural decoding. Finally, the measured field is inverted to the original bitstream by a deep network—e.g. a U-Net of the kind already used for fast acoustic-hologram synthesis [23]. The network *architecture* is mature, but the *inverse* field-to-source decoding ACPCM requires is harder and less established than the forward synthesis those works demonstrate: such inverse problems are ill-posed and a network can hallucinate plausible outputs, so any demonstration must include held-out, information-theoretic controls (Prediction 3).

6 Epistemic ledger

Table 1 is the paper's central honesty device: each subsystem is paired with the real-world mechanism it borrows and with a blunt status label. The pattern is deliberate: *the read path and the write* ***actuation*** *are built from established techniques, while the speculative weight is concentrated in a small number of clearly marked assumptions about how freezing water records and preserves an acoustic field.* The refuted ingredient ("water memory") is load-bearing nowhere; it is motivational framing only.

7 Falsifiable predictions

A speculative proposal earns its keep by being testable. ACPCM implies concrete, disprovable predictions; any one of them failing falsifies the corresponding assumption.

ACPCM subsystem	Borrowed real mechanism	Status of the leap
Rapid "write" by phase change (section 4)	HAMR heat–quench; chalcogenide PCM melt–quench switching [3, 9]	**Speculative** (Assumption 3)
Cell isolation via pods (section 2)	Charge-trap flash discrete traps [5]	**Plausible** (Assumption 1)
Cymatic addressing grid (section 3)	Acoustofluidic patterning; holographic tweezers [20, 17]	Particle nodes **established**; molecular ordering **speculative** (Assumption 2)
Sonification codebook (section 4)	Data-sonification methodology [24, 25]	Mapping **established**; Russell octaves **esoteric/optional** [37]
MHz acoustic-holography read-out (section 5)	Acoustic holograms; hydrophone metrology; DL inversion [12, 22, 23]	**Established** techniques
"Structured-water" carriers (section 2)	EZ/fourth-phase water; Orch-OR water [33, 30]	**Contested / minority** [32, 31]
"Water memory" (motivation only)	Benveniste high-dilution claims [34]	**Refuted** [35]

Table 1. Epistemic ledger mapping each ACPCM subsystem to its borrowed real mechanism and the status of the conceptual leap. The architecture is engineered so that the *refuted* ingredient (water memory) carries *no* load—it is motivational framing only—while the genuinely novel risk is isolated in Assumption 3. Established citations support only their own row and lend no credibility to the speculative rows.

Prediction 1 (Field-to-lattice transfer)**.** Freezing a thin water film while broadcasting a fixed MHz pressure pattern produces a *reproducible*, pattern-dependent modulation in the resulting ice (e.g. in density, birefringence, or scattered-field signature) that is absent under sham (unmodulated) freezing. *Test:* interferometric or acoustic readout of patterned vs. control freezes under blinding; *falsified if* no above-noise, pattern-correlated structure survives (Assumption 3).

Prediction 2 (Addressable nodes)**.** The post-freeze structure is spatially organized at the wavelength scale of the imposed standing wave. *Falsified if* the frozen modulation shows no correlation with the node geometry (Assumption 2).

Prediction 3 (Decodable channel)**.** A network trained on (pattern → scattered-field) pairs recovers the input code above chance on held-out patterns, with a measurable channel capacity. *Falsified if* decoding accuracy is at chance on held-out data, indicating no recoverable information (and guarding against the network merely memorizing or hallucinating).

If Prediction 1 fails—the most likely outcome given current physics—the architecture collapses to its established components, and that negative result is itself worth reporting.

8 Discussion: obstacles, limits, and why this is hard

We do not claim ACPCM is feasible. The honest reading is that it is *probably not*, for reasons worth stating precisely because they sharpen the predictions above.

The freezing transition is violent. Water expands by roughly 9% on freezing and releases substantial latent heat; nucleation is stochastic and growth is dendritic. Any acoustic "imprint" must survive a disordering, mechanically disruptive event: a ~9% volume change (a ~3% linear strain) at a voxel scale of $\lambda/2 \approx$ 15–75 µm (10–50 MHz in water) implies linear displacements of order ~0.5–2 µm—a non-negligible fraction of the intended nodal spacing itself. Assumption 3 is thus in direct tension with the thermodynamics of solidification. Controlling nucleation site, rate, and crystal habit (hexagonal ice I_h vs. vitrified/amorphous ice) is itself a hard, partly unsolved problem.

Liquid water has no long-term memory. The hydrogen-bond network reorganizes on picosecond timescales [29]; there is no credible evidence for persistent information storage in *liquid* water, and the specific "water memory" claims invoked as motivation were investigated and not reproduced [35, 34], including later revivals [36]. ACPCM only sidesteps this objection by storing nothing in the liquid—which places the entire burden on the freeze step.

Resolution versus penetration. Resolving microscopic features acoustically requires very high frequencies, where absorption in water and ice rises steeply; there is a real, physics-imposed tension between resolution and penetration that caps achievable density. Read-out is also an ill-posed inverse problem, and hydrophone metrology already shows tens-of-percent inter-device disagreement for tightly focused MHz fields [22], so any demonstration must include rigorous, blinded, information-theoretic controls (Prediction 3).

Thermodynamic and energy cost. Writing a bit by freezing and reading it by illumination implies an energy budget per bit that is plausibly orders of magnitude worse than incumbent technologies, before considering throughput. Even if every assumption held, the density/energy/latency envelope would need to beat HAMR [3] and 3D NAND [7] to matter—a very high bar.

What would change our mind. A positive, replicated result on Prediction 1 under blinded controls—pattern- dependent, above-noise structure that survives freezing and carries recoverable information—would promote Assumption 3 from speculation to phenomenon and justify pursuing the rest. Absent that, ACPCM remains an organizing metaphor and a teaching device for cross-disciplinary analogy, not a storage technology.

9 Conclusion

We have presented Acousto-Cymatic Phase-Change Memory as a structured thought experiment that transposes two proven storage principles—rapid phase-change writing (HAMR [3], PCM [9]) and microscopic isolation (CTF [5])—onto a rapidly frozen aqueous medium, with an acoustic-holographic, neural-decoded read path that reuses otherwise-established techniques [12, 23]. Its value is not a feasibility claim but the discipline of the synthesis: by laying out an explicit epistemic ledger (table 1) and a falsifiable test program (section 7), the proposal makes clear exactly which single physical claim (Assumption 3) it lives or dies by, concentrates all genuinely speculative risk there, and quarantines refuted ideas as motivation only. We offer it in that spirit: a provocation to be tested and, most likely, falsified—which is the useful kind.

Epistemic disclosure

This manuscript is a speculative "idea paper." Where it invokes "water memory," "structured/EZ water," or W. Russell's octave cosmology, it does so with explicit labels indicating these are contested, minority, or refuted positions, used as motivational or historical framing only; none carries the proposal's logical weight. The established mechanisms cited (HAMR, charge-trap flash, phase-change memory, acoustic holography, deep-learning acoustic inversion, data sonification) are mainstream, and each is cited only to support its own subsystem; their combination into ACPCM is hypothetical and unvalidated. Nothing here should be read as a claim that the device works. All references were independently verified against Crossref, publisher, arXiv, or ADS records during preparation.

Acknowledgements

The synthesis draws on a reading list spanning storage engineering, acoustics, and the history of fringe science; the author thanks the open literature and explicitly credits the critical and skeptical sources [31, 32, 35] that establish the boundaries of accepted knowledge.

References

[1] S. H. Charap, P.-L. Lu, and Y. He, "Thermal stability of recorded information at high densities," *IEEE Trans. Magn.*, vol. 33, no. 1, pp. 978–983, Jan. 1997, doi: 10.1109/20.560142.

[2] D. Weller and A. Moser, "Thermal effect limits in ultrahigh-density magnetic recording," *IEEE Trans. Magn.*, vol. 35, no. 6, pp. 4423–4439, Nov. 1999, doi: 10.1109/20.809134.

[3] M. H. Kryder, E. C. Gage, T. W. McDaniel, W. A. Challener, R. E. Rottmayer, G. Ju, Y.-T. Hsia, and M. F. Erden, "Heat assisted magnetic recording," *Proc. IEEE*, vol. 96, no. 11, pp. 1810–1835, Nov. 2008, doi: 10.1109/JPROC.2008.2004315.

[4] W. A. Challener *et al.*, "Heat-assisted magnetic recording by a near-field transducer with efficient optical energy transfer," *Nat. Photonics*, vol. 3, no. 4, pp. 220–224, 2009, doi: 10.1038/nphoton.2009.26.

[5] M. H. White, D. A. Adams, and J. Bu, "On the go with SONOS," *IEEE Circuits Devices Mag.*, vol. 16, no. 4, pp. 22–31, Jul. 2000, doi: 10.1109/101.857747.

[6] H. Tanaka *et al.*, "Bit cost scalable technology with punch and plug process for ultra high density flash memory," in *2007 IEEE Symp. VLSI Technology, Tech. Dig.*, Jun. 2007, pp. 14–15, doi: 10.1109/VLSIT.2007.4339708.

[7] A. Goda, "3-D NAND technology achievements and future scaling perspectives," *IEEE Trans. Electron Devices*, vol. 67, no. 4, pp. 1373–1381, Apr. 2020, doi: 10.1109/TED.2020.2968079.

[8] S. R. Ovshinsky, "Reversible electrical switching phenomena in disordered structures," *Phys. Rev. Lett.*, vol. 21, no. 20, pp. 1450–1453, 1968, doi: 10.1103/PhysRevLett.21.1450.

[9] M. Wuttig and N. Yamada, "Phase-change materials for rewriteable data storage," *Nat. Mater.*, vol. 6, no. 11, pp. 824–832, 2007, doi: 10.1038/nmat2009.

[10] S. Raoux, W. Wełnic, and D. Ielmini, "Phase change materials and their application to nonvolatile memories," *Chem. Rev.*, vol. 110, no. 1, pp. 240–267, 2010, doi: 10.1021/cr900040x.

[11] M. Faraday, "On a peculiar class of acoustical figures; and on certain forms assumed by groups of particles upon vibrating elastic surfaces," *Phil. Trans. R. Soc. Lond.*, vol. 121, pp. 299–340, 1831, doi: 10.1098/rstl.1831.0018.

[12] K. Melde, A. G. Mark, T. Qiu, and P. Fischer, "Holograms for acoustics," *Nature*, vol. 537, no. 7621, pp. 518–522, 2016, doi: 10.1038/nature19755.

[13] G. Memoli, M. Caleap, M. Asakawa, D. R. Sahoo, B. W. Drinkwater, and S. Subramanian, "Metamaterial bricks and quantization of meta-surfaces," *Nat. Commun.*, vol. 8, art. 14608, 2017, doi: 10.1038/ncomms14608.

[14] S. Jiménez-Gambín, N. Jiménez, J. M. Benlloch, and F. Camarena, "Holograms to focus arbitrary ultrasonic fields through the skull," *Phys. Rev. Applied*, vol. 12, art. 014016, 2019, doi: 10.1103/PhysRevApplied.12.014016.

[15] M. D. Brown, "Phase and amplitude modulation with acoustic holograms," *Appl. Phys. Lett.*, vol. 115, no. 5, art. 053701, 2019, doi: 10.1063/1.5110673.

[16] M. D. Brown, B. T. Cox, and B. E. Treeby, “Stackable acoustic holograms,” *Appl. Phys. Lett.*, vol. 116, no. 26, art. 261901, 2020, doi: 10.1063/5.0009829.

[17] A. Marzo, S. A. Seah, B. W. Drinkwater, D. R. Sahoo, B. Long, and S. Subramanian, “Holographic acoustic elements for manipulation of levitated objects,” *Nat. Commun.*, vol. 6, art. 8661, 2015, doi: 10.1038/ncomms9661.

[18] A. Marzo, M. Caleap, and B. W. Drinkwater, “Acoustic virtual vortices with tunable orbital angular momentum for trapping of Mie particles,” *Phys. Rev. Lett.*, vol. 120, art. 044301, 2018, doi: 10.1103/PhysRevLett.120.044301.

[19] D. J. Collins, B. Morahan, J. Garcia-Bustos, C. Doerig, M. Plebanski, and A. Neild, “Two-dimensional single-cell patterning with one cell per well driven by surface acoustic waves,” *Nat. Commun.*, vol. 6, art. 8686, 2015, doi: 10.1038/ncomms9686.

[20] J. Rufo, F. Cai, J. Friend, M. Wiklund, and T. J. Huang, “Acoustofluidics for biomedical applications,” *Nat. Rev. Methods Primers*, vol. 2, art. 30, 2022, doi: 10.1038/s43586-022-00109-7.

[21] International Electrotechnical Commission, *IEC 62127-1:2022, Ultrasonics — Hydrophones — Part 1: Measurement and characterization of medical ultrasonic fields*, 2nd ed. Geneva: IEC, 2022.

[22] E. Martin and B. E. Treeby, “Investigation of the repeatability and reproducibility of hydrophone measurements of medical ultrasound fields,” *J. Acoust. Soc. Am.*, vol. 145, no. 3, pp. 1270–1282, 2019, doi: 10.1121/1.5093306.

[23] Q. Lin, J. Wang, F. Cai, R. Zhang, D. Zhao, X. Xia, J. Wang, and H. Zheng, “A deep learning approach for the fast generation of acoustic holograms,” *J. Acoust. Soc. Am.*, vol. 149, no. 4, pp. 2312–2322, 2021, doi: 10.1121/10.0003959.

[24] G. Kramer, B. Walker, T. Bonebright, P. Cook, J. H. Flowers, N. Miner, and J. Neuhoff, *Sonification Report: Status of the Field and Research Agenda*. Report prepared for the U.S. National Science Foundation by members of the Int. Community for Auditory Display (ICAD), 1999. [Online]. Available: https://digitalcommons.unl.edu/psychfacpub/444/

[25] T. Hermann, A. Hunt, and J. G. Neuhoff, Eds., *The Sonification Handbook*. Berlin: Logos Verlag, 2011, ISBN 978-3-8325-2819-5. [Online]. Available: https://sonification.de/handbook/

[26] T. Hermann, “Taxonomy and definitions for sonification and auditory display,” in *Proc. 14th Int. Conf. Auditory Display (ICAD)*, Paris, France, Jun. 2008.

[27] J. W. Trayford, C. M. Harrison, R. C. Hinz, M. Kavanagh Blatt, S. Dougherty, and A. Girdhar, “Inspecting spectra with sound: proof-of-concept and extension to datacubes,” *RAS Tech. Instrum.*, vol. 2, no. 1, pp. 387–392, 2023, doi: 10.1093/rasti/rzad021.

[28] W. Smith, “Interactive musical periodic table: sonification of visible element emission spectra,” in *Proc. 21st Sound and Music Computing Conf. (SMC)*, 2024.

[29] M. L. Cowan, B. D. Bruner, N. Huse, J. R. Dwyer, B. Chugh, E. T. J. Nibbering, T. Elsaesser, and R. J. D. Miller, “Ultrafast memory loss and energy redistribution in the hydrogen bond network of liquid H_2O,” *Nature*, vol. 434, pp. 199–202, 2005, doi: 10.1038/nature03383.

[30] S. Hameroff and R. Penrose, “Consciousness in the universe: a review of the ‘Orch OR’ theory,” *Phys. Life Rev.*, vol. 11, no. 1, pp. 39–78, 2014, doi: 10.1016/j.plrev.2013.08.002. [Contested hypothesis.]

[31] M. Tegmark, “Importance of quantum decoherence in brain processes,” *Phys. Rev. E*, vol. 61, no. 4, pp. 4194–4206, 2000, doi: 10.1103/PhysRevE.61.4194. [Skeptical rebuttal to microtubule quantum coherence.]

[32] D. C. Elton, P. D. Spencer, J. D. Riches, and E. D. Williams, “Exclusion zone phenomena in water—a critical review of experimental findings and theories,” *Int. J. Mol. Sci.*, vol. 21, no. 14, art. 5041, 2020, doi: 10.3390/ijms21145041. [Critical review of the “fourth phase” interpretation.]

[33] G. H. Pollack, *The Fourth Phase of Water: Beyond Solid, Liquid, and Vapor*. Seattle, WA: Ebner & Sons, 2013, ISBN 978-0-9626895-4-3. [Self-published; contested minority hypothesis—cite with [32].]

[34] E. Davenas, F. Beauvais, J. Amara, *et al.* (J. Benveniste), “Human basophil degranulation triggered by very dilute antiserum against IgE,” *Nature*, vol. 333, pp. 816–818, 1988, doi: 10.1038/333816a0. [Original “memory of water” claim; discredited—see [35].]

[35] J. Maddox, J. Randi, and W. W. Stewart, "'High-dilution' experiments a delusion," *Nature*, vol. 334, no. 6180, pp. 287–290, 1988, doi: 10.1038/334287a0. [On-site investigation: effect did not survive blinding.]

[36] L. Montagnier, J. Aïssa, S. Ferris, J.-L. Montagnier, and C. Lavallée, "Electromagnetic signals are produced by aqueous nanostructures derived from bacterial DNA sequences," *Interdiscip. Sci. Comput. Life Sci.*, vol. 1, no. 2, pp. 81–90, 2009, doi: 10.1007/s12539-009-0036-7. [Non-reproduced fringe revival.]

[37] W. Russell, *The Universal One*. New York: privately published by the author, 1926. [Esoteric/occult cosmology; historical primary source for the octave/spiral metaphor only, not a chemistry reference. Facsimile: https://archive.org/details/the-universal-one-1926-walter-russell.]